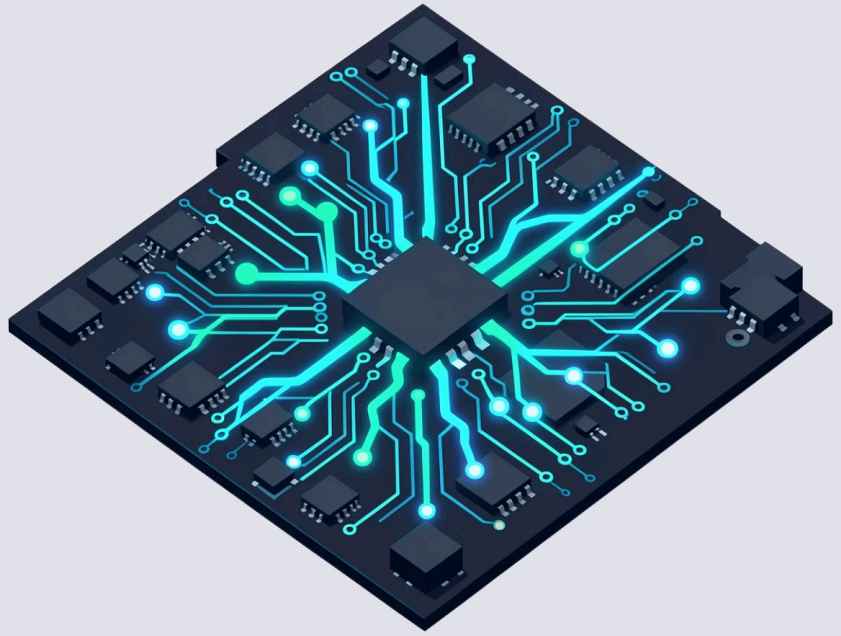
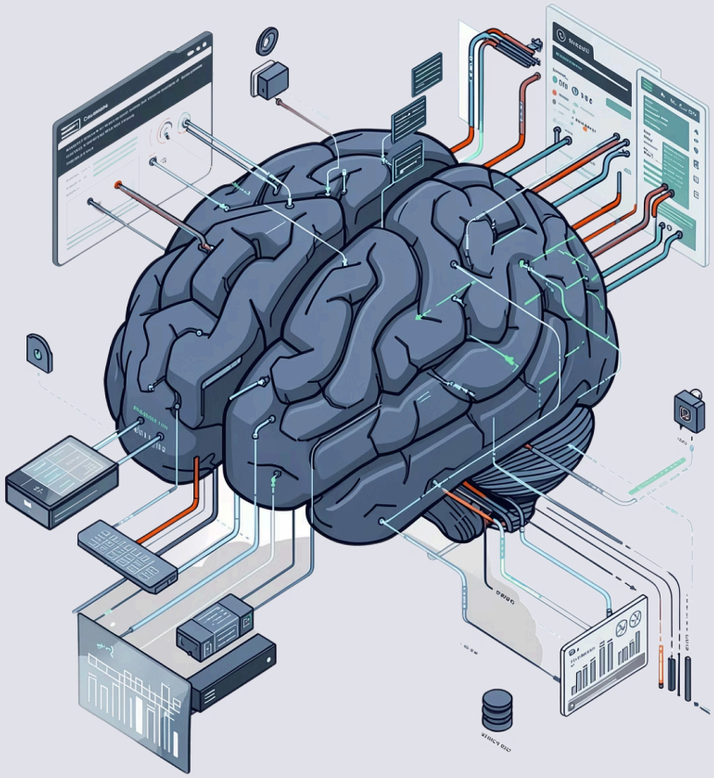
Edge AI and On-Device Learning

Understanding ML Model Deployment on Edge Devices, Challenges & Strategies for Real-Time Analytics

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What is Artificial Intelligence (AI)? — A Quick Recap

🤖 Artificial Intelligence (AI)

Teaching machines to think and make decisions like humans

📊 Machine Learning (ML)

A type of AI where machines learn from data — no manual programming needed

🧠 Deep Learning

ML using layers of "neurons" (like the human brain) to recognize patterns

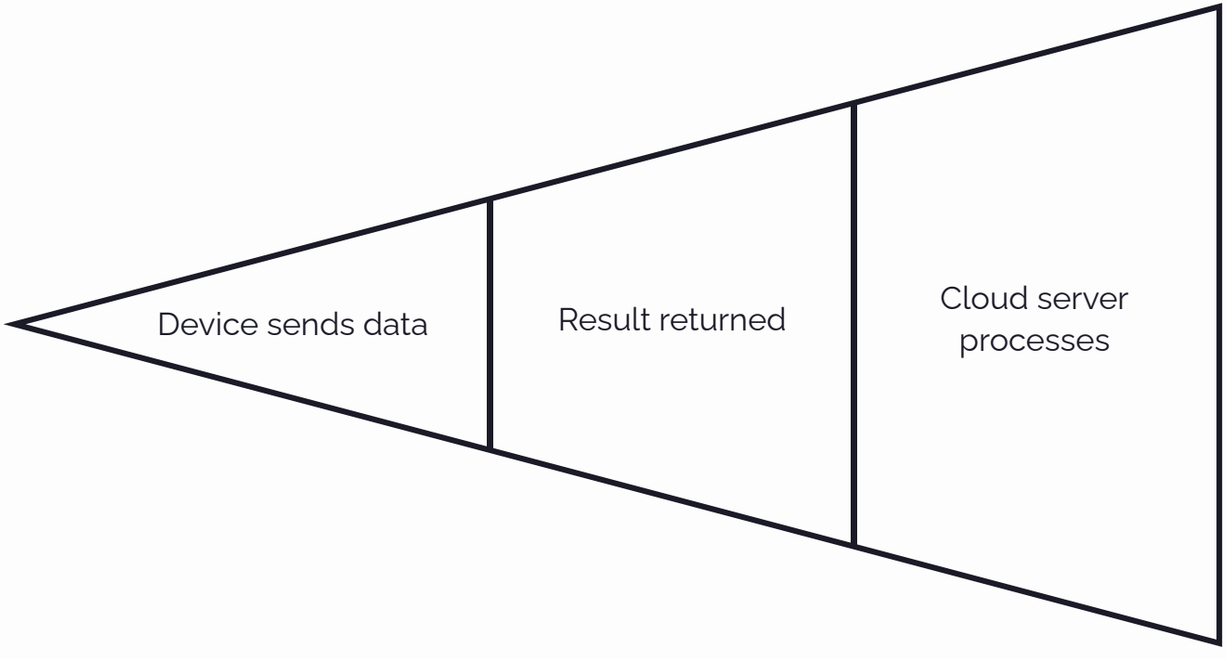
☁️ Traditional AI

Runs on powerful cloud servers far away from the user

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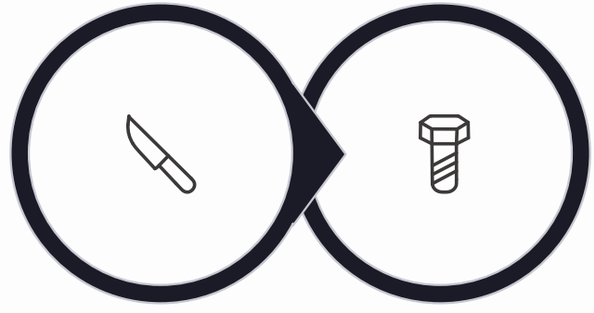
Cloud AI vs. Edge AI — What's the Difference?

☁️ Cloud AI



Data travels from your device → Internet → Cloud Server → Result comes back. **Slow, needs internet.**

⚡ Edge AI



AI runs directly **ON your device** — no internet needed, instant results.

> 💡 **Simple Analogy:** Cloud AI = calling a friend for advice | Edge AI = knowing the answer yourself

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What is Edge AI? — Definition & Key Idea



Definition

Edge AI = Running ML models on or near the device that generates data, instead of a remote cloud.

Edge Devices Include:

- Smartphones & smartwatches
- Cameras & sensors
- Microcontrollers & IoT gadgets

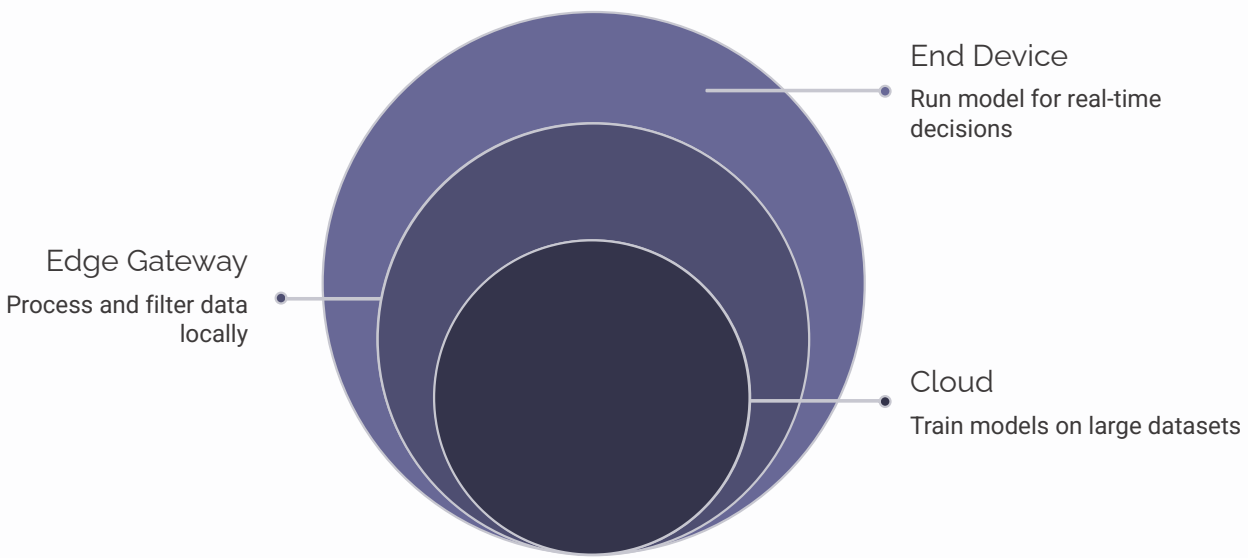
Why Now?

Devices are getting **smarter, cheaper, and more powerful** every year — making on-device AI practical and affordable.

⚡ Faster	🔒 Private	📶 Offline

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Edge AI Architecture — How It All Fits Together



Layer 1 — ☁️ Cloud

Train the ML model using **large datasets** on powerful servers. This is where the heavy lifting happens.

Layer 2 — 🔄 Edge Gateway

Intermediate device (e.g., Raspberry Pi, industrial gateway) that **processes and filters** data locally before passing it on.

Layer 3 — 📟 End Device

Tiny sensor or microcontroller runs the final model for **real-time decisions** at the point of action.

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What is On-Device Learning?

🔵 Inference Only

Device uses a **fixed pre-trained model** — no learning happens on the device. Fast and simple.

🟢 On-Device Training

Device **updates the model** using its own local data — learns and improves over time without sending data to the cloud.

Example: Your phone's keyboard learns your typing style over time — without ever sending your messages to the cloud. That's on-device learning in action!

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How ML Models Are Deployed on Edge Devices — Step by Step



01

Train

Build & train the ML model on a powerful cloud/PC using large datasets

02

Compress

Shrink the model (quantization, pruning) so it fits on a small device

03

Convert

Convert to edge-friendly format (TensorFlow Lite, ONNX)

04

Deploy

Load the model onto the edge device (phone, microcontroller, camera)

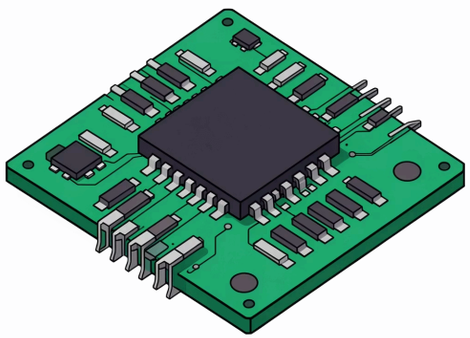
05

Infer

Device runs the model on live data — real-time results!

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Challenges of Edge AI — Why Is It Hard?



⚡ Limited Power

Edge devices run on tiny batteries — AI is power-hungry and drains them fast

🧠 Limited Memory

Microcontrollers may have less than 1 MB RAM — ML models can be GBs in size

🐢 Limited Processing Speed

No powerful GPU — computations are slow compared to cloud servers

📡 No/Unreliable Internet

Must work offline in remote or disconnected areas

🔒 Security & Privacy

Protecting local data from attacks and breaches on resource-constrained devices

Optimization Strategy 1 — Quantization

The Problem

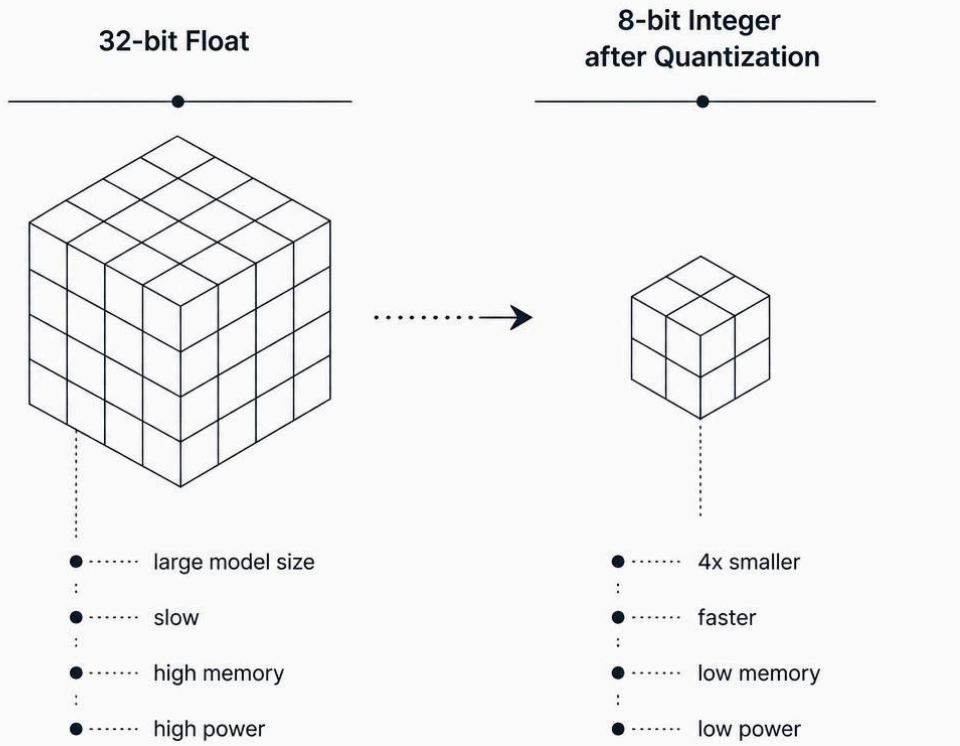
ML models store numbers as **32-bit floating point values** — very large, memory-hungry, and slow on edge hardware.

The Solution

Quantization reduces numerical precision from 32-bit floats → 8-bit integers. This cuts memory traffic and lets inference run on simpler arithmetic units.

The Result

...with **minimal accuracy loss**.



4× Smaller	Runs Faster	Less Power

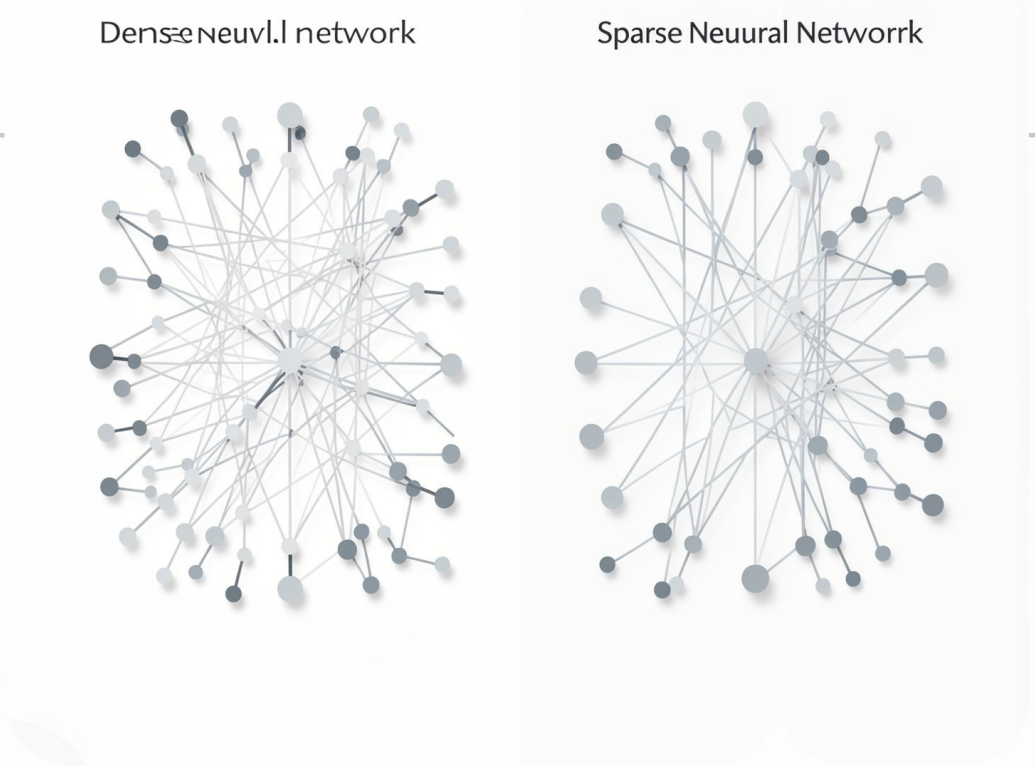
📷 **Analogy:** Like compressing a photo from RAW to JPEG — smaller file, still looks great!

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Optimization Strategy 2 — Pruning & Knowledge Distillation

Before Pruning

Dense network with many redundant connections



After Pruning

Sparser network with removed low-importance weights

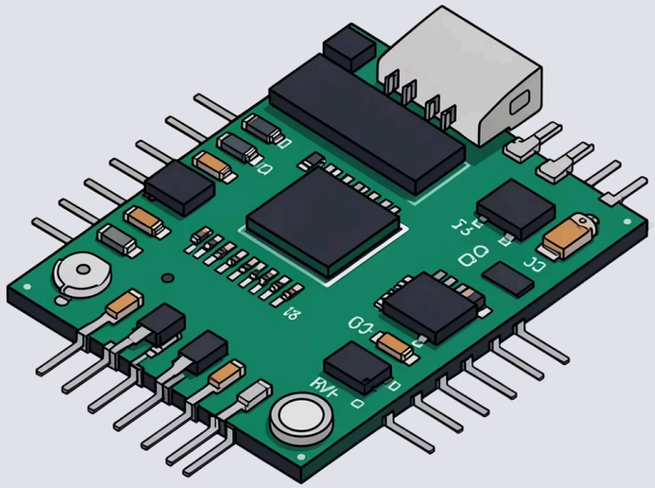
✂️ Pruning

Remove unnecessary "weights" (connections) in the neural network that contribute little to accuracy. Like **trimming dead branches from a tree** — the tree still grows, but is lighter and healthier.

🎓 Knowledge Distillation

Train a small **"student" model** to copy the behavior of a large **"teacher" model**. Result: a tiny model that performs almost as well as the big one — perfect for edge devices!

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Optimization Strategy 3 — TinyML & Efficient Architectures

TinyML

Running ML on ultra-low-power microcontrollers — designed to run for **years on a single coin-cell battery**. A whole field dedicated to squeezing AI into the tiniest devices!

Efficient Model Designs

MobileNet, SqueezeNet — built small from the start, not shrunk later. Engineered specifically for resource-constrained environments.

Deployment Tools

TensorFlow Lite, Edge Impulse, ONNX Runtime — purpose-built frameworks for converting and running models efficiently on edge hardware.

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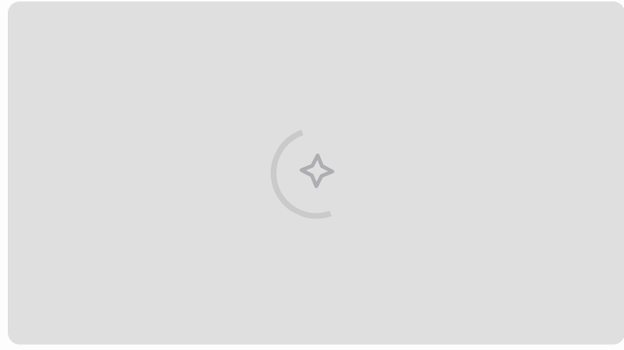
Federated Learning — Learning Together Without Sharing Data

The Problem

To improve AI, we need more data — but sharing personal data is a serious privacy risk.

The Solution

Federated Learning: Each device trains locally on its own data, then only shares **model updates** (not raw data) with the server. The server combines all updates to improve the global model.



Real-World Example

Google Keyboard improves for everyone — without reading anyone's private messages. 🔒

✅ Privacy preserved. ✅ AI improves. ✅ No raw data leaves your device.

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Real-World Applications of Edge AI



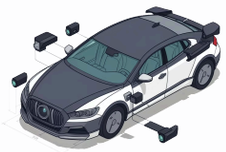
📱 Smartphones

Face unlock, voice assistants (Siri/Google), real-time photo enhancement — all running on-device



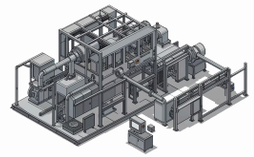
🏥 Healthcare

Wearables detecting heart anomalies, fall detection for the elderly — life-saving real-time AI



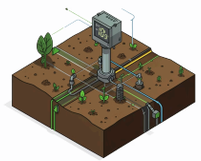
🚗 Autonomous Vehicles

Real-time object detection, lane keeping — no cloud delay allowed when driving at speed



🏭 Industry 4.0

Predictive maintenance — detecting machine faults before they happen, reducing downtime



🌾 Agriculture

Soil sensors + AI for smart irrigation decisions — works even with no internet in rural areas

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Summary — Edge AI at a Glance

✅ What is Edge AI?

AI that runs on the device itself, not the cloud — local, fast, and private

✅ Benefits

Low latency, privacy protection, works offline, enables real-time decisions

✅ Challenges

Limited memory, power, and processing speed on edge devices

✅ Solutions

Quantization, Pruning, TinyML, Federated Learning

✅ Tools

TensorFlow Lite, Edge Impulse, ONNX Runtime

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Key Takeaways & What's Next?

🔑 Key Takeaways

The Future is Edge

Billions of smart devices will run AI locally